

Abstract

In this study, a physics-guided data augmentation framework was developed for predicting the yield strength of Friction Stir Welded joints using machine learning techniques. Response Surface Methodology (RSM) and Latin Hypercube Sampling (LHS) were used to generate a physically consistent augmented dataset of nearly 550 samples. In addition to the primary welding parameters, physics-based features such as Heat Input, Rotational Pitch, and Tool Peripheral Velocity were incorporated to improve model interpretability and prediction capability.

Five machine learning models — Linear Regression, Random Forest, Gradient Boosting, XGBoost, and Artificial Neural Network — were trained and evaluated using R^2 , RMSE, and MAE metrics. Among all models, Gradient Boosting achieved the best performance with the highest prediction accuracy and lowest error. Correlation analysis, residual analysis, and sensitivity analysis further confirmed that the developed framework successfully captured the thermo-mechanical behavior of the FSW process.

The results demonstrate that combining physics-guided feature engineering with machine learning can provide an efficient and reliable approach for weld strength prediction while reducing dependence on extensive experimental trials.

Introduction

Friction stir welding (FSW) was born out of necessity. For decades, engineers had sought a reliable way to join aluminium alloys – prized for their light weight and strength – into strong structures. But aluminium proved surprisingly hard to weld by conventional fusion methods. It conducts heat about five times faster than steel, and its surface forms a tenacious aluminium-oxide layer with a much higher melting point than the base metal [1]. In practice, arc or fusion welding often caused problems: the heat would dissipate too quickly, leading to incomplete fusion or tiny voids. The oxide layer could trap inclusions and cracks could form easily. As a result, welded aluminium parts often had poor strength or needed costly rework. These challenges limited the use of aluminium in critical aerospace and automotive structures.

In 1991, researcher Wayne Thomas and others conceived Friction Stir Welding [2] instead of melting the metal, FSW uses a spinning, non-consumable tool. A rotating pin and shoulder plunge between two plates, generating frictional heat that softens the aluminium just enough to stir and forge it together under pressure. The metal never becomes liquid. This “solid-state” joining avoids many issues of fusion welding: it prevents solidification cracks and yields a fine-grained, tough weld zone [2]. FSW can produce high-strength, defect-free welds in all aluminium alloys and even join

dissimilar metals. In effect, FSW fulfilled the long-sought promise of welding aluminium without its traditional headaches.

By 1999, Boeing had applied FSW to build the interstage of a Delta II rocket the first launch with major friction-stirred aluminium components [3]. A few years later, NASA used FSW to fabricate the large external fuel tanks for its space shuttles, launching a set of three tanks in 2001 [3]. The key advantages were clear: friction stir welds of aluminium had high tensile strength, low distortion, and minimal porosity compared to traditional welds. By mixing the material under pressure below the melt point, FSW yielded very uniform and refined grain structures across the joint.

Early trials uncovered new kinds of defects. If the tool was fed too slowly or too quickly, “wormhole” or tunnel defects could appear at the weld center. A particularly insidious flaw called a kissing bond or lack-of-penetration zone sometimes formed along the joint interface when material didn’t fully fuse. Other issues included occasional surface cracks, excessive flash (squeezed-out metal), and asymmetric microstructures. Tool wear also emerged as a challenge when welding very hard or abrasive aluminium alloys.

A slight tilt angle on the tool (usually 2–3 degrees) was introduced so that the shoulder brushes material toward the weld, forging it together [4]. Concave shoulder profiles were used to pull plasticized metal inward. These changes in tool geometry and positioning greatly improved metal flow and consistency [4].

As the practice of FSW matured, computational methods and optimization tools entered the picture. Engineers employed design of experiments (DOE) and response surface methods (RSM) to systematically study how weld speed, tool rotation, and other factors affect strength and quality. These techniques helped build empirical models that guided parameter choices. More recently, machine learning (ML) approaches have been applied to friction stir welding data. Researchers have trained neural networks, support vector machines, random forests and other algorithms to predict outcomes like tensile strength or hardness from weld inputs. For example, one study used a DOE-generated dataset to train several regression models (linear, kernel-SVM, random forest, etc.) for an aluminium FSW joint. They found that an optimized random forest regressor gave the most accurate strength predictions [5]. Reviews of the field note that artificial neural networks (ANN) are popular for FSW modelling, often combined with genetic algorithms or fuzzy logic to tweak the model or the data. These AI-driven studies demonstrate that data-driven models can fit FSW results very well, in some cases within just a few MPa of error.

However, several challenges still exist in current data-driven approaches for friction stir welding. One major limitation is the lack of physical interpretability in many machine learning models. Most existing models act as “black-box” systems that can predict weld

strength with good accuracy but fail to clearly explain the physical reasons behind the prediction. In many studies, process parameters such as rotational speed and welding speed are directly used as model inputs without considering the actual thermal and material flow phenomena occurring during welding. As a result, the influence of heat generation, frictional interaction, and plastic material deformation on weld quality is not properly captured.

Because of this limitation, engineers often still rely on experimental observations and physical intuition to understand why certain welding conditions produce stronger or weaker joints. Therefore, although existing machine learning approaches provide accurate predictions, they often lack transparency and physical meaning. Most current FSW prediction studies focus more on numerical accuracy than on establishing a clear connection between welding physics and mechanical behaviour.

To address these gaps, our work takes a physics-guided approach to data and machine learning. Instead of feeding raw process parameters alone to the models, we construct meaningful features that encode the underlying welding physics. Specifically, we focus on (1) Heat Input – the energy per unit length, which scales with tool torque and rotation speed over travel speed; (2) Weld Pitch – the distance the tool travels per revolution (travel speed divided by rotation speed), which captures how much the tool “steps” along the joint each turn; and (3) Tool Linear Velocity – the tangential speed at the tool edge (π times diameter times rotation speed), which relates to how fast the tool surface moves through the metal. These features together represent the thermal and stirring action in a more direct way than raw speed numbers.

At the same time, we bolster our dataset through a physics-consistent augmentation. Using response surface methodology (RSM) and Latin Hypercube Sampling (LHS), we generate additional synthetic weld conditions that respect realistic ranges of speed, rotation, and tool sizes. This fills in gaps between the original experiments and helps the machine learning models see a broad range of “what-if” scenarios. We then train and compare multiple regression models – from simple linear regression (LR) to ensemble methods like Random Forest (RF) and XGBoost, a kernelized Support Vector Regressor (SVR), and an ANN. By evaluating all these models on both raw and augmented data, we can see not only which technique predicts yield strength best, but also whether they consistently emphasize the same physics-based features.

In the following sections, we will first describe our data collection and augmentation process, then the details of each machine learning model and its performance. Finally, we interpret the results in terms of microstructure and metallurgy. For example, we expect the models to reveal that a higher weld pitch (fast travel or slow rotation) tends to increase yield strength by limiting peak temperature and refining the grain structure, and that moderate heat input (not too high) gives the toughest joints. These

interpretations will link back to well-known ideas like the Hall–Petch effect (finer grains strengthen the metal). In this way, our approach not only predicts outcomes but also connects them to the physics of friction stir welding. By the end of the paper, readers will see how a physics-informed ML model can guide weld parameter choices and provide insight beyond what a black-box predictor alone can offer.

Methodology

The methodology adopted in this study focused on developing a physics-guided data augmentation framework for predicting the yield strength of Friction Stir Welded (FSW) joints. The complete process was carried out in multiple stages, beginning with experimental data collection and ending with the preparation of a validated augmented dataset for machine learning applications.

1. Experimental Data Collection and Preprocessing

The initial dataset was prepared using experimentally obtained Friction Stir Welding process parameters along with their corresponding yield strength values. Three major welding parameters were selected as input variables:

- Shoulder Diameter (D)
- Welding Speed (V)
- Tool Rotational Speed (N)

The output response considered for the study was Yield Strength (MPa). Before using the dataset for augmentation and modelling, the experimental data was carefully preprocessed. Inconsistent values, duplicate entries, and physically unrealistic parameter combinations were removed to improve the reliability of the dataset.

In addition, feasible operating ranges for the welding parameters were defined based on practical FSW conditions reported in literature and industrial practices. This preprocessing stage ensured that all further generated samples remained physically meaningful and manufacturable under actual welding conditions.

2. Development of Response Surface Model (RSM)

To establish the relationship between welding parameters and yield strength, a Response Surface Methodology (RSM)-based regression model was developed. RSM was selected because of its ability to model nonlinear interactions between process parameters efficiently.

A second-order quadratic regression equation was used to represent the relationship between input variables and output response. The general form of the model is given below:

$$Y = \beta_0 + \sum \beta_i x_i + \sum \beta_{ij} x_i x_j + \sum \beta_{ii} x_i^2$$

where:

- (Y) represents the yield strength,
- (x_i) represents the welding process parameters,
- and (β) denotes the regression coefficients.

The developed RSM model enabled prediction of yield strength for new parameter combinations without performing additional physical experiments. This model later served as the basis for generating synthetic data samples required for machine learning applications.

3. DOE-Based Synthetic Data Generation Using Latin Hypercube Sampling

After establishing the response surface model, a Design of Experiments (DOE)-based data augmentation approach was implemented using Latin Hypercube Sampling (LHS). The purpose of this stage was to generate additional process parameter combinations while maintaining uniform coverage of the design space.

The parameter ranges selected for sampling were:

Parameter	Range
Shoulder Diameter (D)	10–20 mm
Welding Speed (V)	100–200 mm/min
Tool Rotational Speed (N)	600–1000 RPM

LHS was preferred over random sampling because it ensures better statistical representation of the input space and minimizes clustering of generated samples. As a result, the generated dataset contained a wider and more balanced distribution of welding conditions.

The newly generated parameter combinations were then passed through the trained RSM model to estimate corresponding yield strength values. Using this approach, approximately 550 synthetic samples were generated, significantly increasing the size of the original dataset while preserving physical consistency.

4. Physics-Guided Feature Engineering

To further improve the interpretability and predictive capability of the machine learning models, additional physics-based features were derived from the original welding parameters. These engineered features were selected because they directly represent the thermo-mechanical behaviour occurring during the FSW process.

Heat Input

Heat input was calculated using:

$$HI = \frac{N \times D^3}{V}$$

This parameter represents the amount of thermal energy generated during welding and strongly influences material softening and microstructural evolution.

Rotational Pitch

Rotational pitch was determined using:

$$Pitch = \frac{V}{N}$$

This feature describes the distance travelled by the tool during one complete revolution and affects material flow characteristics and heat distribution.

Tool Peripheral Velocity

The peripheral velocity of the tool was calculated as:

$$U = \pi DN$$

This parameter represents the linear velocity of the rotating tool surface responsible for frictional heating and plastic deformation.

The inclusion of these physics-based features helped the machine learning models capture the underlying welding mechanisms more effectively rather than relying only on raw process parameters.

5. Validation of the Augmented Dataset

Finally, the generated synthetic dataset was validated using physical and process-based constraints to ensure the reliability of the augmented data. The validation process confirmed that:

- All generated process parameters remained within realistic welding limits,
- Predicted yield strength values were physically meaningful,
- and the generated samples reflected actual Friction Stir Welding behaviour.

After successful validation, the final augmented dataset was utilized for training and evaluating machine learning models for yield strength prediction. This physics-guided augmentation approach helped improve dataset diversity while maintaining the physical integrity of the welding process.

6. Machine Learning Model Training

After the preparation and validation of the augmented dataset, machine learning models were trained to predict the yield strength of Friction Stir Welded joints. The final dataset contained approximately 550 samples generated through the physics-guided data augmentation framework.

An 80:20 train-test split was adopted, where 80% of the data was used for training the models and the remaining 20% was used for testing and performance evaluation. Both the original process parameters and the derived physics-based features were used as input variables during training. The input features included:

- Shoulder Diameter (D)
- Welding Speed (V)
- Tool Rotational Speed (N)
- Heat Input
- Rotational Pitch
- Tool Peripheral Velocity

To evaluate the effectiveness of different learning approaches, five machine learning models were trained and compared:

1. Linear Regression (LR)

Used as a baseline statistical model because of its simplicity and interpretability.

2. Random Forest Regression (RF)

Selected for its ability to handle nonlinear relationships and reduce overfitting through ensemble learning.

3. Gradient Boosting Regression (GB)

Implemented to improve prediction accuracy using sequential learning and error minimization.

4. **Extreme Gradient Boosting (XGBoost)**

Used because of its optimized boosting mechanism, computational efficiency, and strong predictive capability for nonlinear datasets.

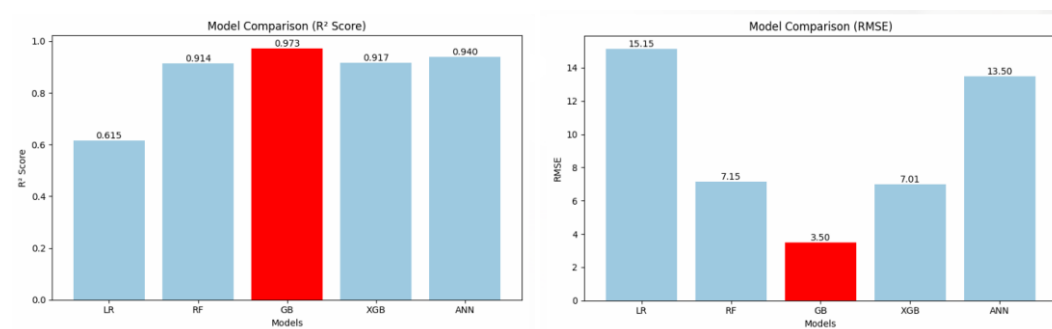
5. **Artificial Neural Network (ANN)**

Applied to capture complex nonlinear interactions among welding parameters and physics-based features.

The performance of these models was later evaluated using statistical performance metrics such as coefficient of determination (R^2), Root Mean Square Error (RMSE), and Mean Absolute Error (MAE) to identify the most suitable model for yield strength prediction.

Results and Discussion

The performance of the developed machine learning models was evaluated using statistical metrics such as coefficient of determination (R^2), Root Mean Square Error (RMSE), and Mean Absolute Error (MAE). These metrics were used to compare the predictive capability of different models for estimating the yield strength of Friction Stir Welded joints.



Graph 1(i)(ii) Model Comparison

The comparative results of all trained models are presented in **Graph 1**, which illustrates the variation in (R^2) score and RMSE values among the different machine learning approaches.

1. **Comparative Performance of Machine Learning Models**

Among all the developed models, the **Gradient Boosting (GB)** model achieved the best overall predictive performance. The model produced the highest coefficient of determination with an (R^2) score of approximately 0.973 and the lowest RMSE value of nearly 3.50 MPa. These results indicate that the Gradient Boosting model was able to

capture the nonlinear relationship between welding parameters and yield strength more effectively than the other models.

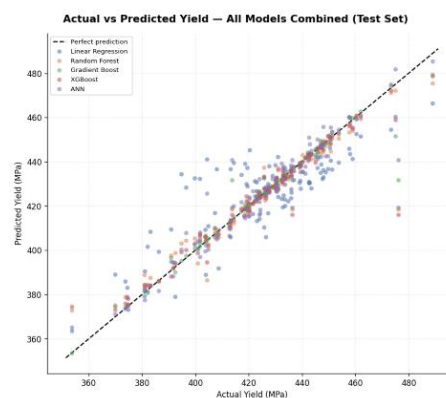
The **Artificial Neural Network (ANN)** also demonstrated strong predictive capability with an (R^2) score close to 0.940. However, its RMSE value remained relatively high compared to Gradient Boosting, suggesting that although ANN captured general trends effectively, it showed larger prediction deviations for certain samples.

The **Random Forest (RF)** and **XGBoost (XGB)** models produced comparable results, with (R^2) scores around 0.91–0.92 and RMSE values near 7 MPa. Both ensemble-based models performed significantly better than Linear Regression because of their ability to model nonlinear interactions among process parameters and physics-based features.

In contrast, **Linear Regression (LR)** exhibited the lowest performance, with an (R^2) score of approximately 0.615 and the highest RMSE value of around 15.15 MPa. The poor performance of Linear Regression indicates that the relationship between FSW parameters and yield strength is highly nonlinear and cannot be accurately represented using a simple linear model.

Overall, the results clearly demonstrate that ensemble boosting techniques, particularly Gradient Boosting, are highly effective for predicting the mechanical behaviour of Friction Stir Welded joints using augmented datasets and physics-informed features.

2. Analysis of Actual vs Predicted Values



Graph 2 Actual vs Predicted Values

The prediction accuracy of the developed models was further examined using **Graph 2**, which presents the Actual vs Predicted scatter plots for each model.

For the Gradient Boosting model, the predicted values were closely distributed around the ideal 45° reference line, indicating strong agreement between experimental and predicted yield strength values. Random Forest and XGBoost also showed relatively good alignment with the reference line, although a slightly larger spread was observed.

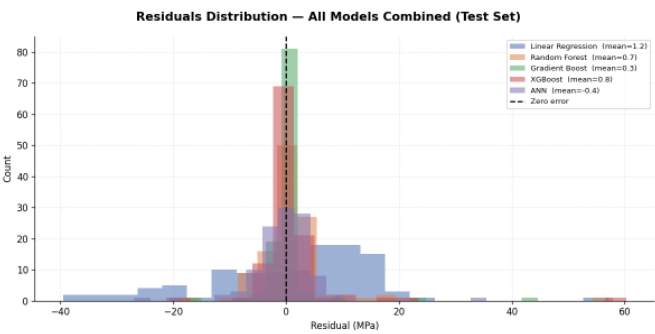
The Linear Regression model displayed significant deviations from the ideal trend, confirming its inability to model the complex nonlinear behavior of the welding process. The ANN model demonstrated good prediction capability but exhibited a few scattered points at higher yield strength values, which contributed to its comparatively higher RMSE.

These observations confirm that Gradient Boosting provided the most consistent and reliable predictions among all models.

3. Residual Error Analysis

Residual analysis was carried out to evaluate the distribution and randomness of prediction errors.

Residual Distribution

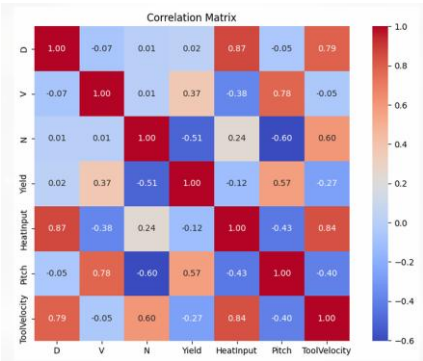


Graph 3 Residual Distribution

Graph 3 illustrates the residual distribution for the developed models. The Gradient Boosting model showed a narrow and symmetric residual distribution centered near zero, indicating minimal prediction bias and stable model behavior.

Random Forest and XGBoost also exhibited reasonably balanced residual distributions, while Linear Regression showed wider residual spread, suggesting larger prediction inconsistencies. ANN residuals displayed moderate variability due to occasional over- and under-predictions.

4. Correlation Matrix Analysis



Graph 4 Correlation Matrix

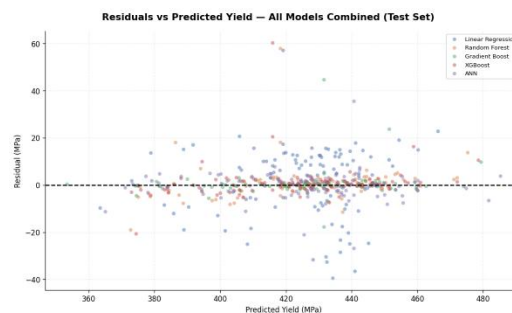
A correlation matrix was generated to study the relationship between process parameters, physics-based features, and yield strength.

The analysis showed that **Welding Speed (V)** had a moderate positive correlation with yield strength (0.37), while **Tool Rotational Speed (N)** showed a negative correlation (−0.51). Among the engineered features, **Pitch** exhibited a strong positive correlation with yield strength (0.57), indicating its significant influence on weld quality.

Additionally, strong correlations were observed between **Heat Input** and **Shoulder Diameter** (0.87), as well as **Tool Velocity** (0.84), confirming the physical relationship between tool geometry, heat generation, and welding behavior.

Overall, the correlation analysis demonstrated that the physics-guided features effectively captured important thermo-mechanical relationships within the FSW process.

5. Residuals vs Predicted Values



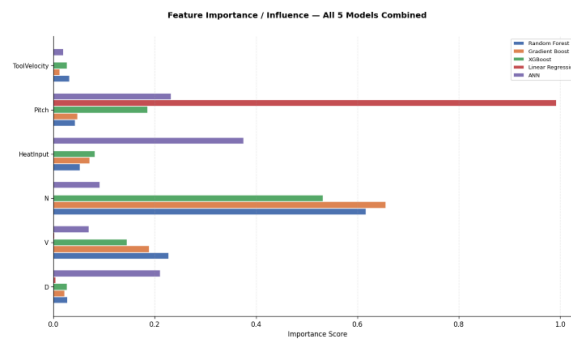
Graph 5

Residuals vs Predicted Values

The residual patterns shown in **Graph 5** further validated model stability. For Gradient Boosting, the residuals were randomly distributed without any visible trend, indicating that the model successfully captured the underlying relationship between input parameters and yield strength.

On the other hand, Linear Regression displayed systematic residual patterns, which implies that important nonlinear relationships were not adequately modeled.

6. Feature Importance Analysis



Graph 6 Feature Importance Analysis

The contribution of individual welding parameters and physics-based features was analyzed using feature importance plots presented in **Graph 6**.

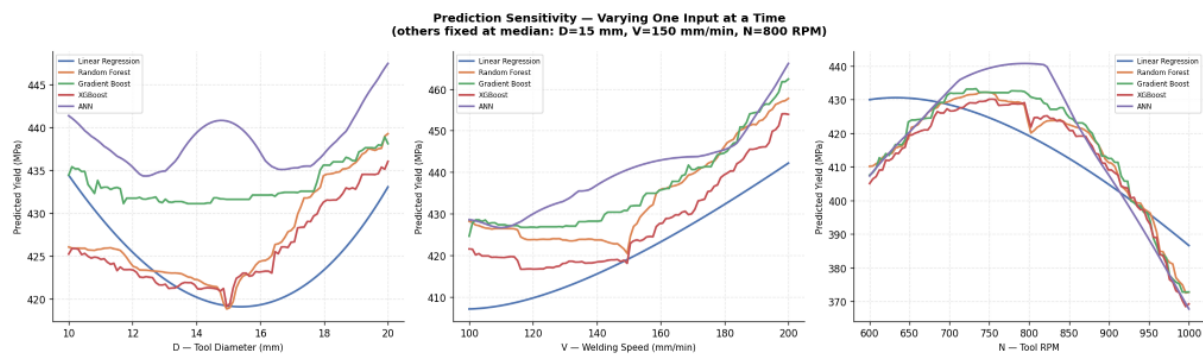
The analysis revealed that physics-guided parameters such as:

- Heat Input
- Rotational Pitch
- Tool Peripheral Velocity

played a major role in determining yield strength. Among them, Heat Input emerged as one of the most influential features because it directly governs thermal energy generation and material softening during welding.

The importance analysis also showed that combining traditional process parameters with derived physical features significantly improved the learning capability of ensemble-based models. This validates the effectiveness of incorporating domain knowledge into the machine learning framework.

7. Prediction Sensitivity Analysis



Graph 7 Prediction Sensitivity Analysis

Sensitivity analysis was performed by varying one process parameter at a time while keeping the others constant. The results are presented in **Graph 7**.

Effect of Shoulder Diameter (D)

Most models showed an initial decrease in yield strength with increasing shoulder diameter, followed by a gradual rise at higher diameters. Gradient Boosting and Random Forest produced smoother and more stable trends, while ANN showed higher fluctuations.

Effect of Welding Speed (V)

The predicted yield strength generally increased with welding speed, especially beyond 150 mm/min. This trend indicates that higher welding speed helps reduce excessive heat exposure and improves joint strength. Ensemble models captured this behavior more effectively than Linear Regression.

Effect of Tool Rotational Speed (N)

All models showed that yield strength increased with rotational speed up to an optimum range of nearly 750–850 RPM and then decreased at higher RPM values. This behaviour is physically consistent with the FSW process, where excessive rotational speed causes overheating and softening of the weld region.

Overall, the sensitivity curves confirmed that the developed machine learning models successfully captured the influence of major FSW process parameters on yield strength prediction.

6. Overall Discussion

The results obtained from this study demonstrate that physics-guided data augmentation combined with ensemble machine learning methods can effectively predict the yield strength of Friction Stir Welded joints.

Among all models, Gradient Boosting provided the best balance of accuracy, robustness, and generalization capability. The inclusion of physics-based engineered features substantially improved prediction performance compared to using only raw process parameters.

The study also highlights that integrating Design of Experiments (DOE), Response Surface Modelling (RSM), and machine learning creates a reliable framework for reducing experimental effort while maintaining predictive accuracy in advanced manufacturing applications.

Conclusion

In this study, a physics-guided data augmentation framework was developed for predicting the yield strength of Friction Stir Welded joints using machine learning techniques. The combination of Response Surface Methodology (RSM), Design of Experiments (DOE), and Latin Hypercube Sampling (LHS) helped generate a reliable augmented dataset while maintaining physical consistency of the welding process.

To improve model interpretability and prediction capability, physics-based features such as Heat Input, Rotational Pitch, and Tool Peripheral Velocity were incorporated along with the original welding parameters. The augmented dataset was then used to train multiple machine learning models including Linear Regression, Random Forest, Gradient Boosting, XGBoost, and Artificial Neural Network.

Among all the models, the Gradient Boosting model achieved the best predictive performance with the highest (R^2) score and the lowest RMSE value, demonstrating its strong capability in capturing the nonlinear relationships present in the FSW process. The results also showed that ensemble-based learning methods performed significantly better than conventional linear regression models.

The correlation analysis, residual analysis, and sensitivity analysis further confirmed that the developed models successfully captured the thermo-mechanical behavior of Friction Stir Welding and produced physically meaningful predictions.

Overall, the proposed methodology provides an efficient and reliable approach for yield strength prediction while reducing the need for large-scale experimental trials. The study also highlights the potential of integrating physics-guided feature engineering with machine learning techniques for advanced manufacturing and welding applications.

References

- 1 Aluminum Welding Challenges and Solutions | AMD Machines
<https://amdmachines.com/blog/aluminum-welding-challenges-and-solutions/>
- 2 3 Industrialisation of friction stir welding for aerospace structures - TWI
<https://www.twi-global.com/technical-knowledge/published-papers/industrialisation-of-friction-stir-welding-for-aerospace-structures-december-2001>
- 4 Functions, Designs and Materials of Friction Stir Welding Tools - TWI <https://www.twi-global.com/technical-knowledge/faqs/friction-stir-welding-tools>
- 5 Machine learning approach with various regression models for predicting the ultimate tensile strength of the friction stir welded AA 2050-T8 joints by the K-Fold cross-

validation method - ScienceDirect

<https://www.sciencedirect.com/science/article/abs/pii/S2352492822021274>